[Project 4] Modeling Cost of Quantization

Supporting quantization requires more intricate design decisions than just adding circuits to quantize. How do the costs weigh against the benefits of quantizing to fewer bits?

Milestone 1

Study the cost of supporting quantization.

- What hardware units are required to support quantization?
  - Some quantization schemes you can explore: NVFP4, MXFP8/FP4/INT8, bfloat16.
- Create a compound component to model the area and energy of these quantization units.
- Pick a workload and augment the spec by adding Einsums to represent the quantization.

Milestone 2

Study the impact of quantization for various workload shapes and quantizations.

- Choose an architecture, include your compound component, and evaluate the energy and latency of various workload shapes and quantizations.
- Compare the energy and latency results to the ideal scenario in which quantizations do not have overhead.
- Study the resulting mappings and report any latency and energy bottlenecks.

Milestone 3

Propose and evaluate architectural optimizations. You can explore the following (or come up with your own ideas):

- How many quantization units are required to avoid latency bottlenecks? Does it differ depending on workload shape and quantization types?
- When quantizing to fewer bits, does scaling the shape or your spatial array result in higher throughput?
- Within the same chip area budget, can you resize on-chip memory and/or add additional memory levels to achieve better energy and latency?

For any architecture you come up with, report:

- energy and latency,
- energy breakdown, and
- the usage of architectural components (analyze the mapping).